

Sound Signatures for Geometrical Shapes

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Figure 1: (a): The sound signatures generated according to Bouba-Kiki [Usnadze 1924], Maluma-Takete [Köhler 1967], and our approaches, (b): Envision of sound signatures generated for commercial products.

Abstract

We propose automatic generation of sound signatures — short pseudowords which can be associated with certain features of geometric shapes such as roundness, spikiness, etc. This research is related to sound symbolism, a sub-track of synesthesia where people describe information with different sounds. First, we propose a method which allows us to automatically calculate a measure of roundness of geometric shapes. We verify the obtained results with the numerous user tests conducted over the century by the sound symbolism researchers. Next, we propose a deep learning-based method of new sound signatures generation for different shapes.

Keywords

Sound Symbolism, Sonification of geometry

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1 Introduction

In everyday life we tend to give short, often one-word descriptions to objects and people around us: big, small, round, smooth, spiky, thin, slender, slim, fat, etc. It helps us to memorize the features of the shapes which are important for us. However, we speak different languages, in which these words will obviously sound different. Therefore, it can be useful to come up with universally recognizable sound descriptions of shapes in a way how the language of

icons (e.g., traffic signs) delivers meaning to people with different cultures and languages. This is what we call “sound signatures” of geometric shapes. These sound signatures can be short musical tunes or combinations of consonants and vowels combined into pseudowords that can be recognized in any language.

For at least a century there have been research works on sound symbolism, a sub-track of synesthesia, where people describe information with different sounds. This research proved that people’s intuitive association between geometrical shape and sound is not arbitrary. It is known as Bouba-Kiki effect—pseudowords which can be associated with “roundness” of the shapes regardless of the cultural level and language of the people. We therefore work on two tasks: /1/ how to automatically measure certain features of geometric shapes such as roundness and spikiness, and /2/ how to come up with sound signatures which can be easily recognized.

2 Related Works

The theory of sound symbolism originates from Dimitri Uznadze’s study on the Bouba-Kiki effect, which asked participants to associate words with shapes [Usnadze 1924]. It was found that the word “bouba” is associated with round shapes while the word “kiki” — with spiky shapes. This research was conducted on many people with different cultural levels and speaking different languages. In the follow-up research by Wolfgang Köhler [Köhler 1967], the participants showed a strong tendency to link the jagged-spiky shapes with the word “takete” and the rounded-smooth shapes with the word “maluma”. It also suggested a non-arbitrary relationship between the word sounds and meanings. Subsequent studies expanded on sound symbolism, and thus examined how sounds convey meanings across languages [Hinton et al. 2006]. In exploring sound-shape correspondence, studies found that consonants are more significant than vowels in the Bouba-Kiki effect [Fort et al. 2014], and spectral attributes of sound helped predict roundness and spikiness [Fort and Schwartz 2022]. Also, in [Löbbers et al. 2021] it is well-proved that certain strokes and acute angles in visual shapes are strongly correlated to sound roughness and sharpness.

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In summary, the Bouba-Kiki effects are well-documented across cultures, with vowels, consonants, and pseudoword structures influencing sound perception. Advances in computational analysis of sound attributes further aid in understanding sound “roundness”.

3 Our Approach

The previous Bouba-Kiki and Maluma-Takete research was mostly based on conducting numerous user tests proving associations of these words with the shapes sketched by the researchers. We, however, are interested in proposing a method how “roundness” of the shapes can be somehow automatically computed so that the previously proven shape-word associations will hold. We, therefore, hypothesized that for the geometric shapes interpolated by polylines and polygons, the roundness can be calculated based on the angles between the adjacent lines/polygons, which are known as “crease angles”. We then investigated how to efficiently compute and average such angles to eventually come up with just one number reflecting how round or spiky the shape is. We tuned our computations so that the results previously published in sound symbolism papers (e.g., [Löbbers et al. 2021][Passi and Arun 2024]) could be 100% replicated. This approach allowed us to verify our hypothesis without conducting additional user tests—they were all done before. The examples of the automatically computed word-to-shape associations can be seen in Fig. 1. Here we test Bouba-Kiki and Maluma-Takete sound signatures as well as additional descriptors proposed by other authors. We have also extended our approach to 3D shapes and studied how the number of polygons can affect our calculations. Thus, the cube will remain detected as spiky regardless of the number of polygons interpolating it, while the 3D Stanford Bunny (Fig. 1a) may become classified as a less round object if the number of polygons interpolating its surface is small.

The second task is to understand how many such words should be used for describing different features of the shapes, including the roundness and spikiness. This research can be referred to many centuries of research (starting from Pythagoras) on how many notes should constitute a musical scale (e.g., 5, 7, or 12) and how many notes in total should be used for composing music (e.g., 88 notes of the piano). In addition, while it was claimed that the words Bouba and Kiki can be accepted as shapes descriptors in every language, it is not 100% accurate. For example, the word Kiki is not very comfortable with Chinese language which does not have this “Ki” sound. To be precise, we used a deep learning approach to make a tool that can be used for the evaluation of these sound signatures words. We used the dataset released in [Fort and Schwartz 2022]. It contains 124 pseudoword samples annotated with the “roundness” from many user studies. We applied a deep learning method to process sound spectrogram inputs into the “roundness” scores. The “roundness” of both shapes and sounds are placed in the range (0, 1), where 1 indicates that they are more similar to the round “Bouba”.

As a result, after predicting the “roundness” of a shape, it is mapped to the pseudoword with the closest value as its sound signature. The reverse process, selecting the most similar shape for a given pseudoword, is also feasible. For example, by using the pseudowords from the dataset in [Fort and Schwartz 2022], the spiky shape “kiki” can now be described by another sound signature such as “vevi”. Similarly, for more 2D and 3D shapes, the most relevant pseudowords were selected, such as “kouko” and



Figure 2: Four popular buildings with outlines which are mapped to “kiki”, “kibouki”, “boubaki”, and “bouba”.

“eumeu” as displayed in Fig. 1a. More importantly, the outcomes of our approaches showed strong correspondence to the results from the previous studies. The shapes “kiki”, “takete”, “bouba”, and “maluma” were fed into the shape roundness-predicting algorithm, which produced the results 0.231, 0.197, 0.779, 0.708, respectively. Similarly, the results of their sound were 0.329, 0.362, 0.663, and 0.844, which also proves the accuracy of our approach.

Thus, with the proposed approach we are able to generate not only two polar descriptions: for the object being round or spiky, such as bouba-kiki, maluma-takete, etc. We can also propose in-between descriptions of mostly round objects with some spikes and mostly spiky objects with a little degree of roundness. For example, we can come up with 4 sound signatures: bouba, boubaki, kibouki, kiki for, very round shapes, round shapes with some spikes, spiky shapes with some roundness, and very spiky shapes. These words then will match the computed degrees of roundness of 0.3, 0.59, 0.75, and 0.846, respectively. Fig. 2 illustrates which shapes will match these sound signatures based on the processing of their contours.

4 Conclusion

Following a century long sound symbolism research, we proposed how to automatically compute roundness and spikiness of geometric shapes interpolated or outlined by polylines and polygons which we proved by matching our results with the numerous user tests conducted in the sound symbolism papers. We then proposed how to automatically generate sound signature pseudowords based on the roundness of the shapes so that different requests for number of such words can be satisfied. We expect that this research can find different practical applications, including the problem of giving easily recognizable names to different products, e.g., as in Fig. 1b.

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